

Evaluating Production Bottlenecks

INSIGHTS AND RECOMMENDATIONS THROUGH DOWNTIME ANALYSIS

Key Performance Indicators

Batches Produced

645

Total production runs



Delayed Batches

363

Experienced production delays



Days Lost

22

Total days lost to downtime



Downtime Factors

885

Total contributing instances



Unique Factors

13

Distinct downtime causes



Max Factors / Batch

4

Highest instances in one batch



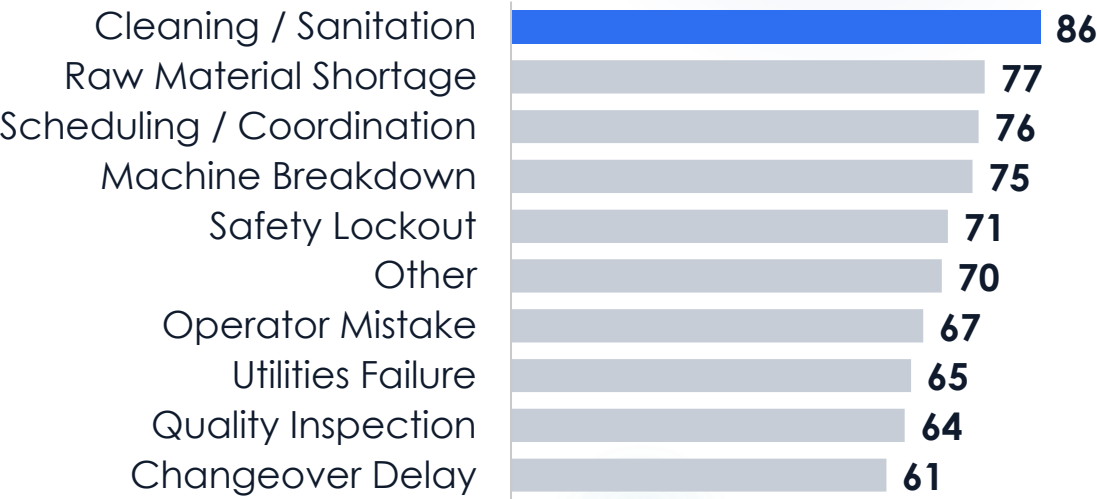
Product Downtime Frequency

Eco-Cleaning Supplies see more downtime than Recyclable Packaging. **GreenFoam Hand Soap** had the most downtimes of any product, driven mainly by raw material shortages and machine breakdowns.

Downtime factor frequency by product



Frequency of downtime factors



Root Cause of Downtime



69%

Non-Operator Factors

Mainly cleaning and sanitation, raw material shortages, and machine breakdowns.



31%

Operator Factors

Mostly scheduling and coordination delays.

Downtime Trends (March–August 2025)

Most downtimes occurred in April and August, with the least in March.

March

Lowest downtime of the period.

April–July

Downward trend, though BioWipe Cleaning Sheets downtime kept rising.

April

Large spike, driven by GreenFoam Hand Soap and RePack Recycled Packaging Film.

August

High downtime again.



Consistent Downtime Causes

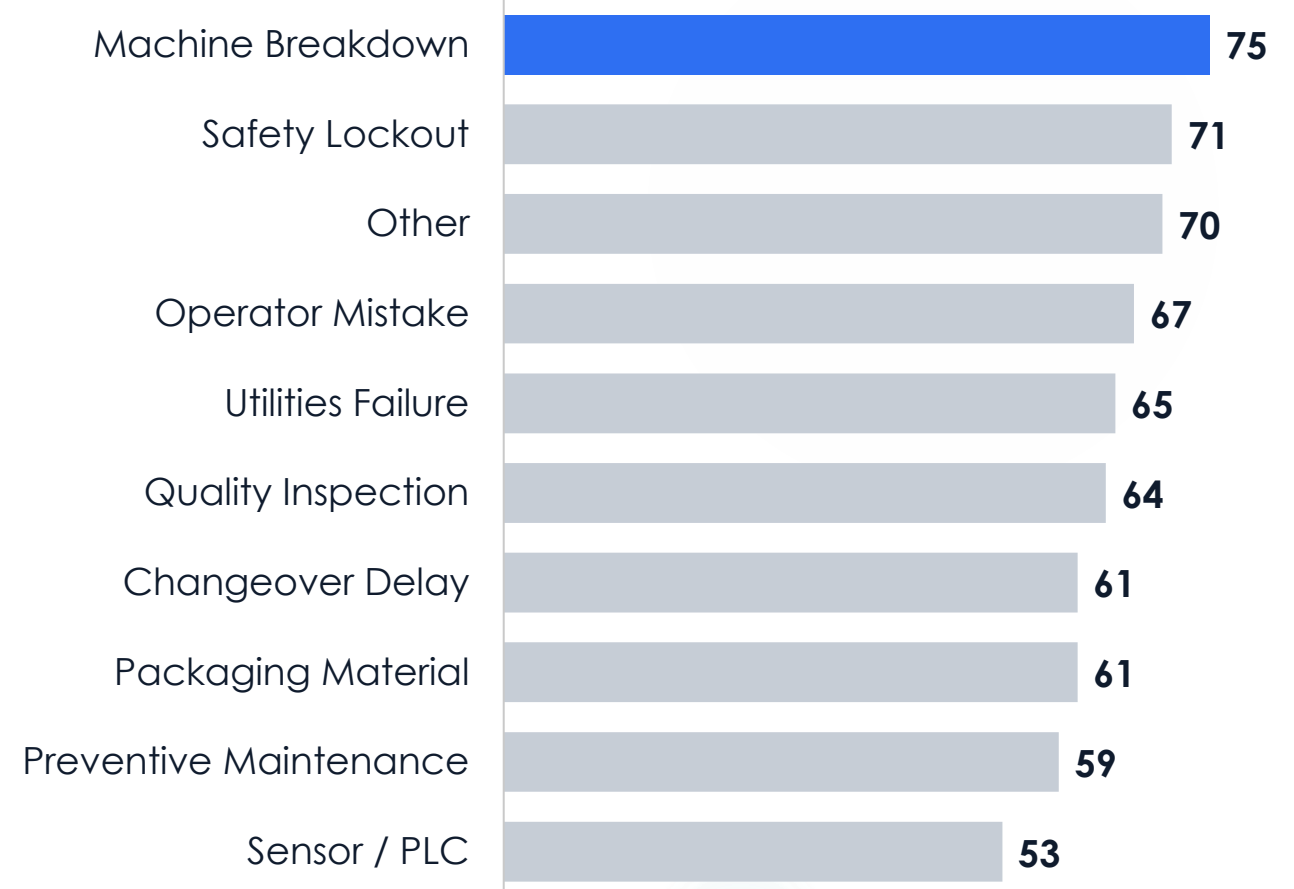
01 Cleaning / Sanitation Cycle

02 Raw Material Shortage

03 Scheduling / Coordination Delay

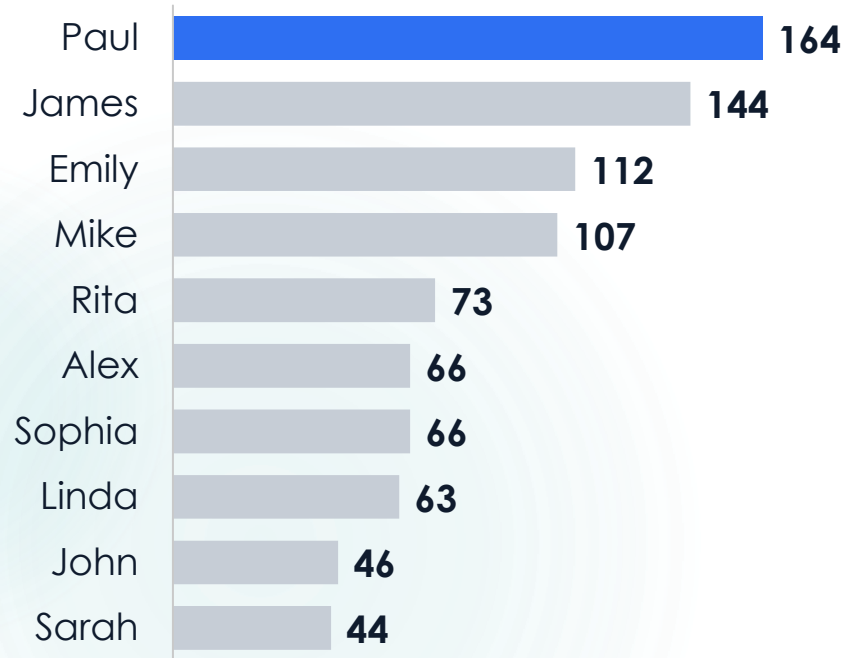
04 Machine Breakdown

Frequency of downtime factors, count of instances

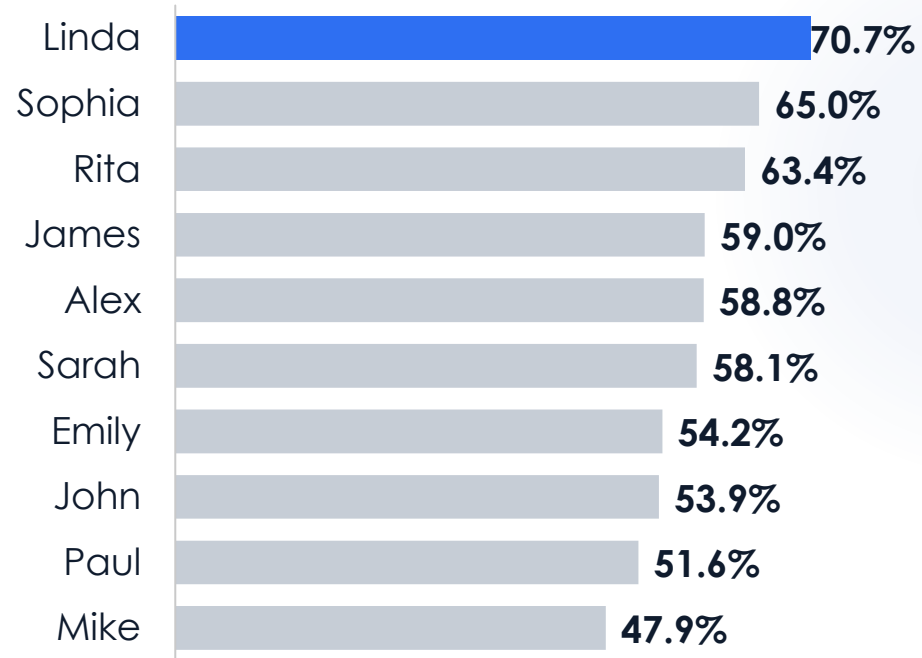


Most Downtimes by Operator

Downtime frequency by operator



Delayed batches, % of batches run



Major factors: safety lockout/emergency stop, machine breakdown, scheduling and coordination delays, cleaning/sanitation, and raw material shortages.

Downtime Impact: Time Lost

Frequency drives cost

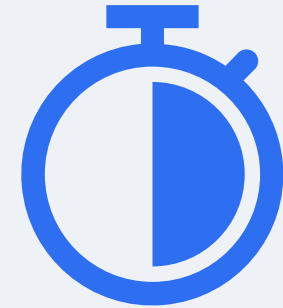
Downtime frequency directly correlates with total time lost.

Primary causes of lost time

Cleaning / sanitation, scheduling / coordination delays, and raw material shortage.

Most time lost

Paul, James and Emily experienced the most time wasted due to downtime.



Scheduling Gaps Identified

The analysis revealed gaps in how operators are scheduled for leading batch production.

Running two or more different products concurrently can be a significant cause of downtime.



Recommendations for Improvement

01

Supply Chain & Cleaning

Improve supply chain coordination and cleaning procedures to prevent delays.

02

Inventory & Alerts

Introduce real-time inventory tracking and automated low raw-material alerts.

03

Preventive Maintenance

Schedule maintenance for high-downtime machines.

04

Batch Scheduling

Implement better tools and production sequencing rules to avoid overlaps.

05

Operator Training & Workload

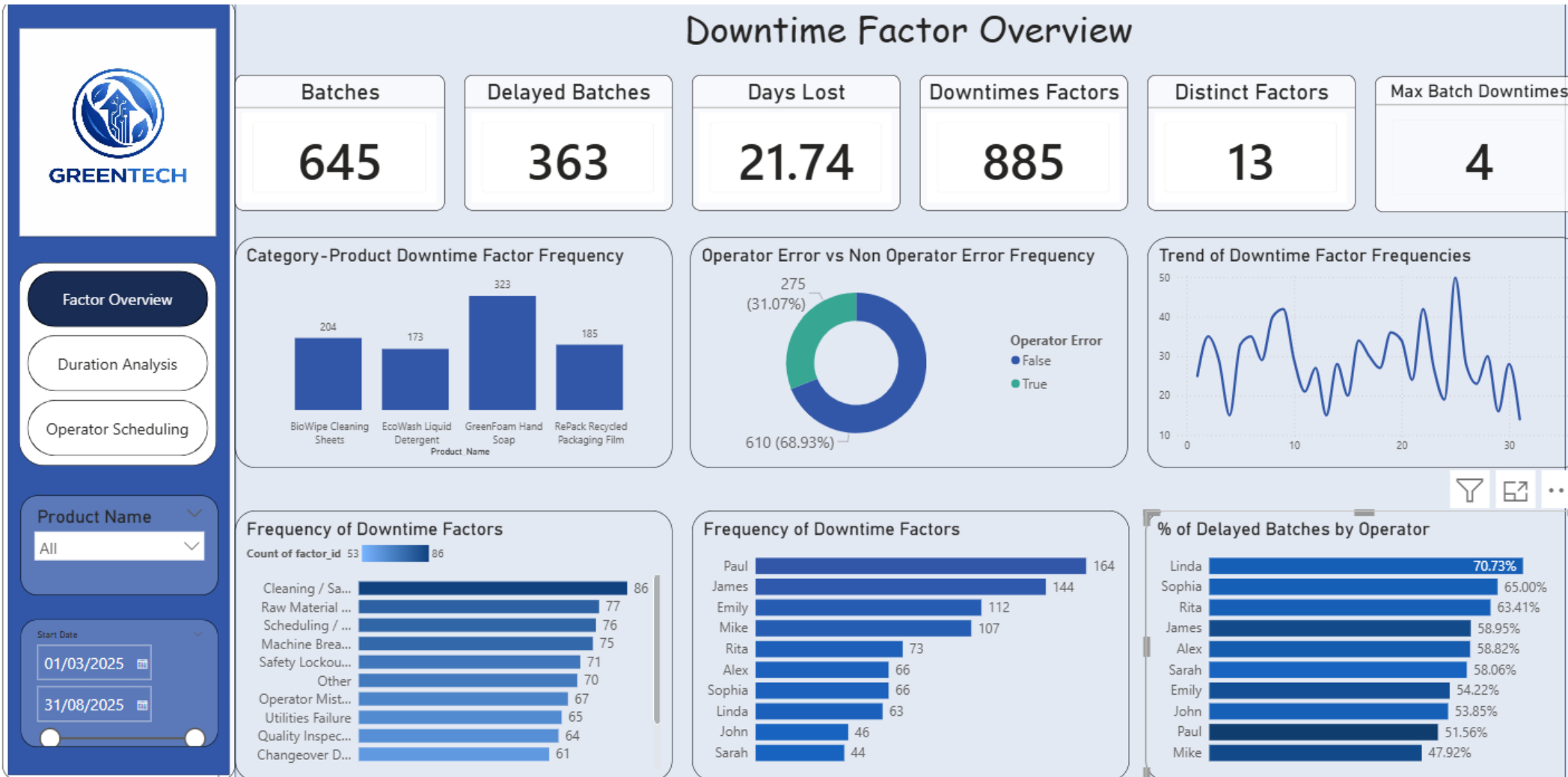
Provide targeted training and review workload allocation for lead operators.

06

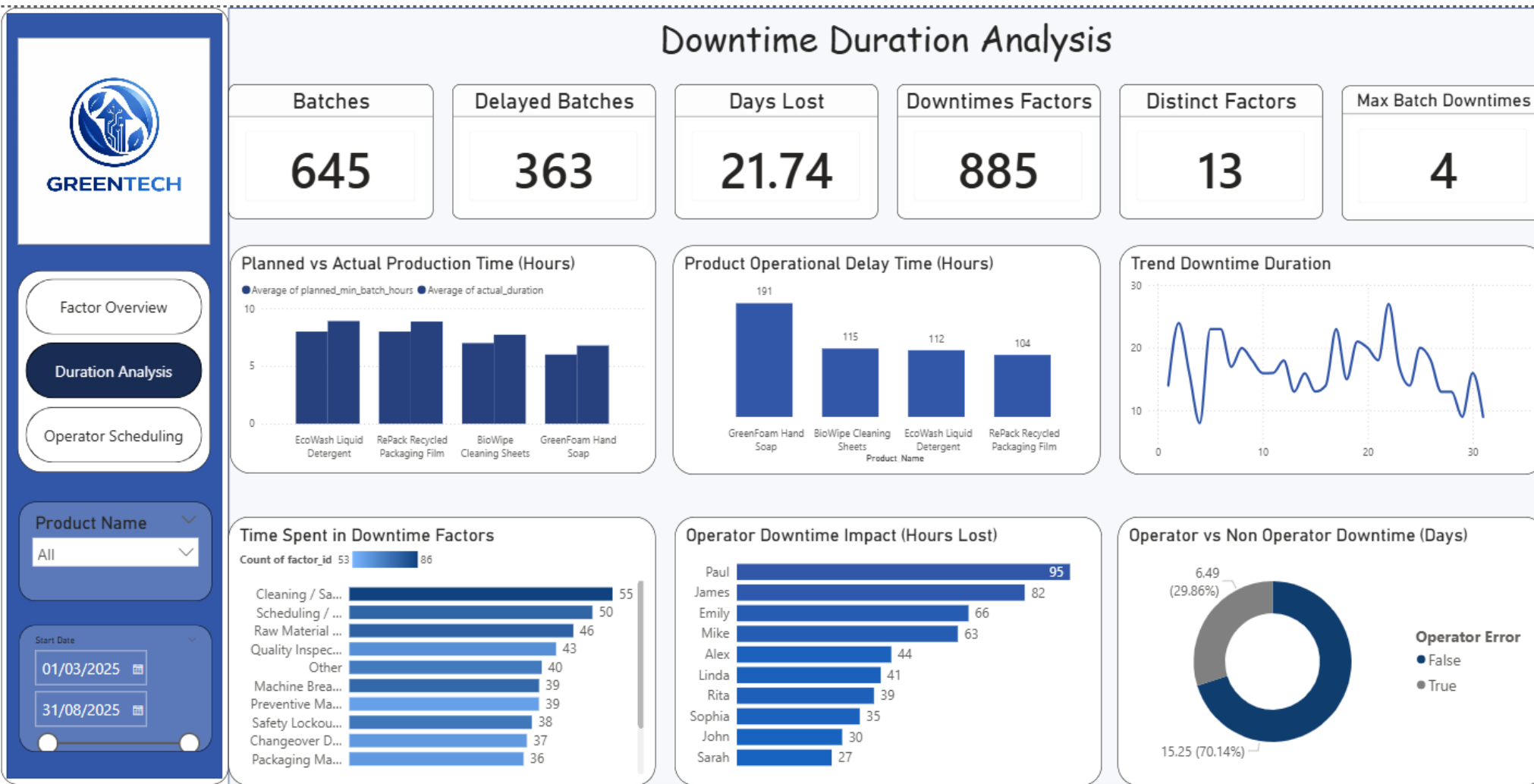
Downtime Reviews

Implement reviews and early-warning systems for proactive action.

Appendix: Downtime Factor Overview



Appendix: Downtime Duration Analysis



Appendix: Operator Scheduling



GREENTECH

Factor Overview

Duration Analysis

Operator Scheduling

Operator

Paul

Start Date

01/03/2025

27/08/2025

Operator Scheduling

Selected Operator

Paul

Days with Multiple Batches

23

Downtime Frequency

52

Batches

49

Daily Multiple Batches Per Day

operator	start_time	Year	Quarter	Month	Day	product_id	Category	Product_Name	batch_id
Paul	06:00:00	2025	Qtr 1	March	1	PRD002	Eco-Cleaning Supplies	BioWipe Cleaning Sheets	272955
Paul	06:00:00	2025	Qtr 1	March	1	PRD003	Eco-Cleaning Supplies	GreenFoam Hand Soap	363707
Paul	06:00:00	2025	Qtr 1	March	4	PRD002	Eco-Cleaning Supplies	BioWipe Cleaning Sheets	248044
Paul	06:00:00	2025	Qtr 1	March	8	PRD002	Eco-Cleaning Supplies	BioWipe Cleaning Sheets	236736
Paul	06:00:00	2025	Qtr 1	March	8	PRD003	Eco-Cleaning Supplies	GreenFoam Hand Soap	397455
Paul	06:00:00	2025	Qtr 1	March	17	PRD003	Eco-Cleaning Supplies	GreenFoam Hand Soap	313343
Paul	13:23:00	2025	Qtr 1	March	17	PRD003	Eco-Cleaning Supplies	GreenFoam Hand Soap	369638
Paul	14:04:00	2025	Qtr 1	March	4	PRD002	Eco-Cleaning Supplies	BioWipe Cleaning Sheets	250774
Paul	06:00:00	2025	Qtr 2	April	16	PRD003	Eco-Cleaning Supplies	GreenFoam Hand Soap	396925
Paul	06:00:00	2025	Qtr 2	May	22	PRD003	Eco-Cleaning Supplies	GreenFoam Hand Soap	396365
Paul	06:00:00	2025	Qtr 2	May	23	PRD003	Eco-Cleaning Supplies	GreenFoam Hand Soap	375429
Paul	06:00:00	2025	Qtr 2	June	5	PRD002	Eco-Cleaning Supplies	BioWipe Cleaning Sheets	270920
Paul	06:00:00	2025	Qtr 2	June	6	PRD002	Eco-Cleaning Supplies	BioWipe Cleaning Sheets	286295
Paul	06:00:00	2025	Qtr 2	June	6	PRD003	Eco-Cleaning Supplies	GreenFoam Hand Soap	371063
Paul	06:00:00	2025	Qtr 2	June	8	PRD003	Eco-Cleaning Supplies	GreenFoam Hand Soap	333911
Paul	06:00:00	2025	Qtr 2	June	10	PRD002	Eco-Cleaning Supplies	BioWipe Cleaning Sheets	239806
Paul	06:00:00	2025	Qtr 2	June	12	PRD002	Eco-Cleaning Supplies	BioWipe Cleaning Sheets	288645
Paul	06:00:00	2025	Qtr 2	June	12	PRD003	Eco-Cleaning Supplies	GreenFoam Hand Soap	378099
Paul	06:00:00	2025	Qtr 2	June	21	PRD002	Eco-Cleaning Supplies	BioWipe Cleaning Sheets	277843